# Statistical Foundations of AI

*Conference Program*

**November 13–14, 2026 • UC Santa Barbara**

Corwin Pavilion and University Center meeting rooms. All times are Pacific Standard Time.

Talks appear in presentation order. Each room's two-hour talk session is divided equally among its speakers. Panel discussions are highlighted.

| Time and room | Friday November 13 |
|---|---|
| **Parallel Session 1** | |
| **1C Corwin**<br>8:30–10:30 am<br>**Statistical computation and structured modeling**<br>Chair: Grace Yi | **Hani Doss** (University of Florida) Theoretical and Empirical Evaluation of a Grouped Gibbs Sampler for Parallel Computation in the Latent Dirichlet Allocation Model<br>**Hao Helen Zhang** (University of Arizona) Dynamic Supervised Principal Component Analysis for High-Dimensional Classification<br>**Yazhen Wang** (University of Wisconsin-Madison) Learning Relationship between Quantum Walks and Underdamped Langevin Dynamics<br>**Ying Nian Wu** (UCLA) Quantum Field Theory as Generative Models, Emergent Gravity as Latent Variable Model<br>**Hongzhe Li** (University of Pennsylvania) Generation-Powered Inference for Distribution-Valued Outcomes<br>**Bertrand Clarke** (U. Nebraska-Lincoln) Uncertainty Quantification via the Posterior Predictive Variance |
| **1F Flying-A**<br>8:30–10:30 am<br>**AI agents and adaptive evaluation**<br>Chair: Jin Zhou | **Jingshen Wang** (UC Berkeley) Agentifying AI Evaluation: What to Evaluate, When to Stop, and How to Estimate<br>**Linjun Zhang** (Rutgers University) Statistical Evaluation of LLMs<br>**David Talby** (John Snow Labs) Statistical design for medical AI safety benchmarks: precision targets, joint balance, and perturbation validity<br>**Jason Klusowski** (Princeton University) Statistical Attribute Alignment for Black-Box Generative AI<br>**Jie Ding** (University of Minnesota) The Future of AI Scientists: Emerging Directions and Fundamental Challenges<br>**Ruoqing Zhu** (University of Illinois at Urbana-Champaign) Building AI Agents for Better Human-Agent Interaction |
| **1H Harbor**<br>8:30–10:30 am<br>**Statistical learning for biomedical data**<br>Chair: Yang Feng | **Jingjing Zou** (UC San Diego) Structured Self-Supervised Learning for Heterogeneous Wearable Data<br>**Andrew Holbrook** (UCLA Biostatistics) Decoding Aging Neurons<br>**Michael Daniels** (Department of Statistics, University of Florida) A case study of causal mediation using Bayesian nonparametrics and semiparametric corrections<br>**Sam Behseta** (Kaiser Permanente Bernard J. Tyson School of Medicine) The Specification Gap: What Artificial Intelligence Leaves Unresolved in Statistical Inference<br>**Lan Xue** (Oregon State University) Cluster Analysis of Longitudinal Associations Between Microbial Abundance and Environmental Factors<br>**Sandra Safo** (University of Minnesota) Scalable Randomized Kernel Methods for Multiview Data Integration and Prediction |
| **1S State Street**<br>8:30–10:30 am<br>**Generative models and learning dynamics**<br>Chair: Jacob Bien | **Ning Ning** (Texas A&M University) Beyond Prediction: World Models for Understanding, Planning, and Action<br>**Yuchen Zhou** (University of Illinois Urbana-Champaign) Optimal Convergence Analysis of DDPM for General Distributions<br>**Shihao Yang** (Georgia Institute of Technology) Statistical Interpretation of AI Architecture and AI Agents for Statistical Research<br>**Zhonghua Liu** (Columbia University) Gradient-Flow Optimization as Dynamic Random-Effects Inference with Applications to Deep Learning<br>**Mingzhao Hu** (Mayo Clinic) ComposeCFM: Flow-Matching Synthetic Generation and Harmonization<br>**Weining Shen** (UC Irvine) SportD: How do VLMs physically strategize? |
| **10:30–11:00 am** | **Break** |

| Time and room | Friday November 13 |
|---|---|
| **11:00–11:05 am**<br>Corwin | **OPENING REMARKS Shelly Gable**<br>Dean of Mathematical, Life, and Physical Sciences at UCSB |
| **11:05 am–12:05 pm**<br>Corwin | **PLENARY TALK Susan Murphy (Harvard University)**<br>Reinforcement Learning for Digital Health Interventions in the Dyadic Setting<br>Chair: Annie Qu |
| **12:05–1:00 pm** | **Lunch** |
| **Parallel Session 2** | |
| **2C Corwin**<br>1:00–3:00 pm<br>**Trustworthy AI and structured inference**<br>Chair: Hua Zhou | **Weijie Su** (University of Pennsylvania) Alignment in Large Language Models: Statistical and Game-Theoretic Perspectives<br>**Aaditya Ramdas** (Stanford University) Controlling the Probability of Unknown Unknowns in Pre-Deployment Testing of AI Systems<br>**Padhraic Smyth** (UC Irvine) Prediction, Confidence, and Calibration in the Age of Deep Learning<br>**Runze Li** (Penn State University) Federated Feature Selection with False Discovery Rate Control<br>**Ji Zhu** (University of Michigan) Bias-Corrected Inference for Edge-Covariate Effects in Latent Space Network Models<br>**Jacob Bien** (University of Southern California) Hierarchical Clustering With Confidence |
| **2F Flying-A**<br>1:00–3:00 pm<br>**Regression and predictive inference**<br>Chair: Yazhen Wang | **Oscar Hernan Madrid Padilla** (UCLA) Risk Bounds For Distributional Regression<br>**Grace Yi** (University of Western Ontario) Residual Correction and Representation Learning for Robust Regression with Continuous Noisy Labels<br>**Boxiang Wang** (University of Iowa) ABC: Certified Selective Evaluation with a Panel of LLM Judges<br>**Matteo Sesia** (University of Southern California) Noise-Adaptive Conformal Classification with Marginal Coverage<br>**Hao Ho/Ker-Chau Li** (UCLA) ATCG: Learning Structural Traits for Auditable Deep Classification<br>**Min Xu** (Rutgers University) Estimation of the sub-Gaussian Parameter |
| **2H Harbor**<br>1:00–3:00 pm<br>**Statistical learning and assisted inference**<br>Chair: Ying Nian Wu | **Kun Chen** (University of Connecticut) Tree-Guided Feature Aggregation as Structured Representation Learning<br>**Emma Jingfei Zhang** (Emory University) Beyond Point Predictions: Distribution-Informed Prediction-Powered Inference<br>**Yang Feng** (New York University) Federated Transfer Learning with Differential Privacy<br>**Kai Zhang** (UNC Chapel Hill) BELIEF in Dependence<br>**Qiyang Han** (Rutgers University) Algorithmic inference via nonconvex gradient descent<br>**Arun Kuchibhotla** (Carnegie Mellon University) Mean-Median gaps in real line and Inference in Infinite-dimensions |
| **2S State Street**<br>1:00–3:00 pm<br>**Network models, representations, and experiments**<br>Chair: Bertrand Clarke | **Xiwei Tang** (University of Texas at Dallas) HODGEDEPTH: SPATIAL REPRESENTATION LEARNING THROUGH GLOBAL GEOMETRY<br>**Keith Levin** (University of Wisconsin—Madison) Testing for correlation between network structure and high-dimensional node covariates<br>**Moulinath Banerjee / Yue Yu** (University of Michigan) Sequential Test Bisection: The Root Finding Problem Revisited<br>**Xiaoyue Maggie Niu** (Pennsylvania State University) A Dyadic Latent Space Model for Directed Networks<br>**Jesús Arroyo** (Texas A&M University) Identifiability and Spectral Estimation in Multilayer Degree-Corrected Mixed Membership Models<br>**Alexander Volfovsky** (Duke University) How to Experiment When You Don’t Yet Know What Matters |
| **3:00–3:30 pm** | **Break** |

| Time and room | Friday November 13 |
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| **Parallel Session 3** | |
| **3C Corwin**<br>3:30–5:30 pm<br>**PANEL DISCUSSION**<br>**Industry perspectives on trustworthy AI**<br>Moderator: Hao Helen Zhang | **PANEL DISCUSSION Industry perspectives on trustworthy AI**<br>**Moderator: Hao Helen Zhang** (University of Arizona)<br>**David Talby** (John Snow Labs)<br>**Sam Behseta** (Kaiser Permanente Bernard J. Tyson School of Medicine)<br>**Guannan Zhai** (Amazon)<br>**Yujia Deng** (Zoom Communications Inc.)<br>**Haoda Fu** (Amgen)<br>**Louis Ehwerhemuepha** (CHOC) |
| **3F Flying-A**<br>3:30–5:30 pm<br>**Online learning and decision making**<br>Chair: Lan Xue | **Xuan Bi** (University of Minnesota) Optimal Auditing of Sequential Decision Systems under Dynamical Constraints<br>**Chengchun Shi** (London School of Economics and Political Science) Demystifying Group Relative Policy Optimization: Its Policy Gradient is a U-Statistic<br>**Yuhua Zhu** (UCLA) Reinforcement Learning in the Physical World<br>**Stefan Wager** (Stanford University) Estimating Dynamic Marginal Policy Effects under Sequential Unconfoundedness<br>**Yufeng Liu** (University of Michigan) Structured Learning of Individualized Decision Rules with Complex Options<br>**Arash Amini** (UCLA) Two Sample Sizes in Test-Time LLM Ensembling: Buffered ERM and Minimax Theory |
| **3H Harbor**<br>3:30–5:30 pm<br>**Statistical inference after learning**<br>Chair: Hongzhe Li | **Lili Zheng** (University of Illinois Urbana - Champaign) LOCO-AdaMP: Built-in LOCO Inference for Adaptive Minipatch Ensembles with Enhanced Prediction<br>**Armeen Taeb** (University of Washington) Consensus Tree Estimation with False Discovery Control via Partially Ordered Sets<br>**Dan Kessler** (UNC Chapel Hill) Inference After Exploration: A Grammar for Adaptive Data Science<br>**Tate Jacobson** (Oregon State University) Global Inference for High-Dimensional Quantile Regression with Heterogeneity<br>**Qi Xu** (University of Minnesota) Towards Efficient Inference under Nonmonotone Missingness with General Imputation<br>**Joshua Cape** (University of Wisconsin-Madison) Network Signflip Parallel Analysis for Selecting the Embedding Dimension |
| **3S State Street**<br>3:30–5:30 pm<br>**Causal discovery and treatment effects**<br>Chair: Hani Doss | **Yuqi Gu** (Columbia University) Discrete Causal Representation Learning<br>**Carlos Cinelli** (University of Washington) Challenges in Causality: Sensitivity Analysis and Automation<br>**Matteo Bonvini** (Rutgers University) On regression with estimated covariates and conditional effects given the propensity score<br>**Yubai Yuan** (The Pennsylvania State University) Causal inference under interference with incomplete exposure mappings<br>**Didong Li** (UNC Chapel Hill) When Can We Trust Synthetic Data for Statistical Inference?<br>**Ying Zhou** (University of Connecticut) Learning from the Unseen: Offline Reinforcement Learning with Hidden Actions |
| **5:30–6:30 pm**<br>Corwin | **Poster session**<br>All posters share this session. See the poster directory. |
| **7:00–8:30 pm**<br>Corwin | **BANQUET TALK Sudipto Banerjee (UCLA)**<br>Carbs, Calories, and Computation: Serving Up Statistical Inference in the AI Ecosystem.<br>Chair: Michele Guindani (UCLA) |

| Time and room | Saturday November 14 |
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| **Parallel Session 4** | |
| **4C Corwin**<br>8:30–10:30 am<br>**Biomedical modeling and AI-assisted inference**<br>Chair: Jin Zhou | **Lily Wang** (George Mason University) Synthetic Surrogate Functional Regression for Neuroimaging Analysis<br>**Lexin Li** (UC Berkeley) Integrating Statistics and AI for Neuroscience Research<br>**Ying Guo** (Emory University) Task-Aware Statistical AI for Learning Directed Brain Connectomes from Neuroimaging Data<br>**Fengzhu Sun** (University of Southern California) Statistical and AI Approaches for Investigating Viruses and Virus-host Interactions<br>**Gang Li** (Department of Biostatistics, UCLA) SAPA: Safe and Adaptive Prediction Augmentation for Survival Analysis<br>**Xinping Cui** (UC Riverside) Scalable Joint Modeling of Multiple Longitudinal Biomarkers and Competing Risks Time-to-Event Data: Applications to Mega-Scale Health Research |
| **4F Flying-A**<br>8:30–10:30 am<br>**Evaluation and inference with AI predictions**<br>Chair: Veronica Berrocal | **Carey Priebe** (Johns Hopkins University) Performance guarantees for LLMs<br>**Junwei Lu** (Harvard University) Language Model Preference Inference via Fisher Random Walk<br>**Will Wei Sun** (Purdue University) Low Rank for Rank: Uncertainty-Aware Task-Specific LLM Ranking from Sparse Pairwise Comparisons<br>**Jiwei Zhao** (University of Wisconsin-Madison) SADA: Safe and Adaptive Aggregation of Multiple Black-Box Predictions in Semi-Supervised Learning<br>**Peng Wang** (University of Cincinnati) Building Statistical AI Tools for Comparative Analysis<br>**Hongquan Xu** (UCLA) Prompt engineering via order-of-addition experiments and screening |
| **4H Harbor**<br>8:30–10:30 am<br>**Structured representations and clustering**<br>Chair: Ali Shojaie | **Shujie Ma** (UC Riverside) Learning What to Share: Adaptive Spectral Integration and Clustering of Multimodal Data<br>**Mladen Kolar** (University of Southern California & MBZUAI) Borrowing Strength Across Studies: Structure-Aware Transfer Learning for High-Dimensional Models<br>**Gongjun Xu** (University of Michigan) Beyond Vintage Rotation: Bias-Free Sparse Representation Learning with Oracle Inference<br>**Jason Xu** (UCLA) Majorization-Minimization Algorithms in Clustering<br>**Xiaohui Chen** (University of Southern California) GEM-KMeans: Memory-Efficient and Accurate Clustering on Massive Scale with GPU Optimization<br>**Marianthi Markatou** (University at Buffalo) A study on kernels for discrete random variables |
| **4S State Street**<br>8:30–10:30 am<br>**Reinforcement learning and adaptive decisions**<br>Chair: Minge Xie | **Feifang Hu** (George Washington University) AI-Powered Adaptive Experiments: From Sequential Analysis to Intelligent Clinical Trials.<br>**Lan Wang** (University of Miami) Quantile Temporal Difference Algorithm for Distributional Reinforcement Learning<br>**Linglong Kong** (University of Alberta) Principled Fast and Meta Knowledge Learners for Continual Reinforcement Learning<br>**Babak Shahbaba** (UCI) Uncertainty-Guided Control: Statistical Foundations for Brain-Inspired Decision-Making<br>**Dehan Kong** (University of Toronto) Successive classification learning for estimating quantile optimal treatment regimes<br>**Walter Dempsey** (University of Michigan) Future-Path Incremental Excursion Effects in Sequentially Randomized Trials |
| **10:30–11:00 am** | **Break** |
| **11:00 am–12:00 pm**<br>Corwin | **PLENARY TALK Xihong Lin (Harvard University)**<br>From Generative AI to Statistical Inference: Harnessing Synthetic Data for Validity and Power<br>Chair: Xiaowu Dai |
| **12:00–1:00 pm** | **Lunch** |

| Time and room | Saturday November 14 |
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| **Parallel Session 5** | |
| **5C Corwin**<br>1:00–3:00 pm<br>**PANEL DISCUSSION**<br>**AI influence and the next generation of research**<br>Moderator: Regina Liu | **PANEL DISCUSSION AI influence and the next generation of research**<br>**Moderator: Regina Liu** (Rutgers University)<br>**George Michailidis** (UCLA)<br>**Bertrand Clarke** (U. Nebraska-Lincoln)<br>**Heping Zhang** (Yale University)<br>**Peter X. K. Song** (University of Michigan)<br>**Radu Craiu** (University of Toronto)<br>**Michael Daniels** (Department of Statistics, University of Florida)<br>**Xiao Wang** (Purdue University)<br>**Ying Wei** (Columbia University)<br>**Michele Guindani** (UCLA) |
| **5F Flying-A**<br>1:00–3:00 pm<br>**Statistical learning and computational inference**<br>Chair: Marianthi Markatou | **Minge Xie** (Rutgers University) Model-Free Inference for High-Dimensional Binary Classification Using Repro Samples<br>**Faming Liang** (Purdue University) An Introduction to Stochastic Deep Learning<br>**Weixin Yao** (UC Riverside) Online Kernel-Based Mode Learning<br>**Ryan Giordano** (UC Berkeley) Black Box Variational Inference with a Deterministic Objective: Faster, More Accurate, and Even More Black Box<br>**Debdeep Pati** (University of Wisconsin-Madison/Boston University) Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests<br>**Ali Shojaie** (University of Washington) GLM Inference with AI-Generated Synthetic Data |
| **5H Harbor**<br>1:00–3:00 pm<br>**Multi-view learning and data integration**<br>Chair: Hua Zhou | **Rong Ma** (Harvard University) Finding Shared Signals Across Many Datasets: Statistical Insights from RMT<br>**Cong Ma** (University of Chicago) Bias-Corrected Subspace Intersection: Minimax-Optimal Shared Subspace Estimation in Multi-View Data<br>**Tiffany Tang** (University of Notre Dame) Consensus dimension reduction via multi-view learning<br>**Ziyi Liang** (UC Santa Barbara) Meta Fusion: A Unified Framework For Multimodal Fusion with Adaptive Mutual Learning<br>**Xinzhou Guo** (HKUST) Stein-Encoder: A White-Box Supervised Encoder via Stein Identities in Multi-Modal Studies<br>**Tian Gu** (Columbia University) Whom to trust and what to keep: adaptive Bayesian transfer learning from pretrained models |
| **5S State Street**<br>1:00–3:00 pm<br>**Statistical learning with heterogeneous data**<br>Chair: Hongtu Zhu | **Zhezhen Jin** (Department of Biostatistics, Columbia University) Subsampling for the analysis of large data<br>**Gourab Mukherjee** (University of Southern California) Kernelized Stein Discrepancy for Learning from Heterogeneous Data<br>**Peijun Sang** (University of Waterloo) Model positive and unlabeled data with a generalized additive density ratio model<br>**Bei Jiang** (University of Alberta) fDA: A data-augmented predictive framework for generating fair synthetic labels with controlled fairness-faithfulness trade-offs<br>**Lu Tang** (University of Pittsburgh) Fairness within Clinical Context: Stratified Tilted Risk Minimization for Medical Prediction<br>**Veronica Berrocal** (UC Irvine) A Bayesian scalar-on-function kernel regression model for crop yield |
| **3:00–3:30 pm** | **Break** |

| Time and room | Saturday November 14 |
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| **Parallel Session 6** | |
| **6C Corwin**<br>3:30–5:30 pm<br>**Reliable learning and structured inference**<br>Chair: Shujie Ma | **Hongtu Zhu** (UNC Chapel Hill) MLFM: Manifold-Based Latent Factor Discovery in Functional Brain Networks<br>**Jian Kang** (University of Michigan) BROWNIE: Bayesian Random Weight Neural Networks as a Simple Alternative to Deep Learning<br>**George Michailidis** (UCLA) Beyond Flat Minima: Sharpness-Aware Bilevel and Minimax Optimization.<br>**Jinchi Lv** (University of Southern California) LIDS: LLM Summary Inference Under the Layered Lens<br>**Tracy Ke** (Harvard University) The Gold Rush in AI4Math: Structural Signatures of AI Involvement<br>**Qing Zhou** (Department of Statistics & Data Science, UCLA) Causal Discovery on Dependent Data |
| **6F Flying-A**<br>3:30–5:30 pm<br>**Foundations of AI and distributed learning**<br>Chair: Faming Liang | **Xiaoqian Liu** (UC Riverside) An Interpretable and Scalable Framework for Evaluating Large Language Models<br>**Nabarun Deb** (University of Chicago Booth School of Business) Sample complexity bounds for categorical Markov random fields via Discrete Diffusions<br>**Wenqing He** (University of Western Ontario) Beyond Marginal Coverage: Efficient Localized Conformal Prediction via Residual Rank Calibration<br>**Rui Miao** (University of Texas at Dallas) TOPPO: Rethinking PPO for Multi-Task Reinforcement Learning with Critic Balancing<br>**Jeffrey Zhang** (UC Riverside) Enhancing Federated Learning Methods using Third Order Optimization<br>**Fei Xue** (Purdue University) Federated Learning for Feature Mismatch |
| **6H Harbor**<br>3:30–5:30 pm<br>**Learning from distributions and complex measurements**<br>Chair: Lily Wang | **Linbo Wang** (University of Toronto) Estimating Wasserstein Barycenters from Sparse Deep Sequencing<br>**Irina Gaynanova** (University of Michigan) Learning from multivariate distributions with nonparanormal transport<br>**Hyunkeun Ryan Cho** (UCSD) Scalable four-level nested functional principal component analysis for large dense functional data<br>**Soham Jana** (University of Notre Dame) Modeling with speckle noise in high-dimensional image sensing<br>**Rui Duan** (Harvard T.H. Chan School of Public Health) Unsupervised Aggregation of Multiple Learning Algorithms<br>**Wen Zhou** (New York University) A Latent Semiparametric Mixture Model for Robust Heterogeneity Learning in High Dimensions |
| **6S State Street**<br>3:30–5:30 pm<br>**Learning from sequential and process data**<br>Chair: Radu Craiu | **Xiucai Ding** (UC Davis) Transformers for Time Series: Statistical Optimality, Structural Exploration and Gradient Descent Consistency<br>**Robert Lund** (UC Santa Cruz) Multiple Changepoint Detection in Non-Gaussian Time Series<br>**Hernando Ombao** (Statistics, Bioengineering, and Applied Mathematics, KAUST) Dependence in Networks: Extremal Vulnerability and Granger Causality Beyond Linear Models<br>**Chunming Zhang** (University of Wisconsin-Madison) Finite-Step Training of Deep ReLU Networks<br>**Zhiliang Ying** (Columbia University) Process data and educational measurement<br>**Yan Yu** (University of Cincinnati) (Mis)Understanding Benign Overfitting in Equity Return Prediction |

# Poster Session

Friday November 13 • 5:30–6:30 pm • Corwin

All posters are presented in this single session. Listings are ordered by the first presenter's surname.

**P01 Susmit Bhattacharyya** (Colorado School of Mines) Marginal Distribution of the LatticeKrig coefficients under Spatial Autoregressive model with Spatial Extremes

**P02 Stevenson Bolivar** (California Polytechnic State University (Cal Poly)) Time-Varying Factor Models for Dynamic High-Dimensional Data

**P03 Keivan Bolouri** (UC Irvine) Aggregation Frequency and Active-Set Agreement in Federated Lasso

**P04 Baiting Chen** (UCLA) Common-agency games for multi-objective test-time alignment

**P05 Xingran Chen** (University of Michigan) A Unified Framework for Inference with General Missingness Patterns and Machine Learning Imputation

**P06 Yifan Chen** (UC Santa Barbara) Statistical Guarantees for Preference Learning with Convex Neural Networks

**P07 Yassine Chouikh and Shannon Li** (UC San Francisco) Verify-or-Abstain Evidence Pipelines: When Cryptographic Custody Is Necessary but Not Sufficient

**P08 Chengyu Cui** (Univ of Michigan - Ann Arbor, MI) Convexity in Disguise: A Theoretical Framework for Nonconvex Low-Rank Matrix Estimation

**P09 Dongyu Cui** (University of Minnesota) Observation-Level Watermarking and Detection for Tabular Data

**P10 Ying Cui** (Stanford University) Learning Interpretable Point-Based Clinical Risk Scores via Direct Optimization

**P11 Annan Deng** (New York University) Geometry-Aware Representation Learning for Functional Connectivity Matrices

**P12 Prasanjit Dubey** (Georgia Institute of Technology) Matching Rates and Optimal Allocation for Federated Probe-Logit Distillation under Heterogeneous Bandwidth Budgets

**P13 Zijun Gao** (USC Marshall) Adaptive sample splitting for randomization tests

**P14 Daeyoung Ham** (University of Texas at San Antonio) Black-Box Knowledge Transfer across Distinct Feature Sets

**P15 Junchao Han** (UC Santa Barbara) Off-Policy Evaluation for Linear Episodic Heterogeneous MDPs by Hierarchical Bellman Residual Minimization

**P16 Rohan Hore** (Stanford University) Conformal Changepoint Localization

**P17 Pingbo Hu** (Columbia University Irving Medical Center) Deep Learning for Continuous-Time Dynamic Survival Prediction With Time-Varying Covariates

**P18 Yichen Hu** (UC Davis) Adaptive Sparsity-Aware Projection-Assisted Spectral Embedding for High-Dimensional Noisy Data with Nonlinear Structures

**P19 Yue Huang** (University of Southern California) iLV-HD: Reconstructing microbial dynamics from high-dimensional compositional time series

**P20 Hyeong Jin Hyun** (University of Cincinnati) Transport-induced Approximate Bayesian Computation

**P21 Ali Kaazempur-Mofrad** (UCLA) Beyond a Single Answer: Proper Scoring of LLM Uncertainty

**P22 Changwoo Lee** (University of Florida) Transformers Can Learn Posterior Predictive Distributions In-Context

**P23 Julianna Lee** (Westat) Using Large Language Models to extract productivity measures from CVs for Program Evaluation

**P24 Chunyin Lei** (UC Santa Barbara) Risk-Aware Quantile Learning for Personalized Dynamic Treatment Regimes

**P25 Dongyang Li** (Department of Mathematics, UC Santa Barbara.) t-Muon: Tensorizing Muon with the t-SVD for Language Model Pretraining

**P26 Pengtao Li** (University of Southern California) Smoothed estimation of Wasserstein barycenters

**P27 Xiaoke Li** (Department of Statistics and Applied Probability, UC Santa Barbara) History-Aware Conformal Prediction for Irregularly Observed Longitudinal Data

**P28 Ling Liang** (The University of Tennessee, Knoxville) A Provably Convergent and Practical Algorithm for Gromov-Wasserstein Optimal Transport

**P29 Qifeng Lin** (UC Davis) Learning common structures from high-dimensional noisy biviews using cross-integral operator

**P30 Yangyang Liu** (Department of Quantitative and Computational Biology, University of Southern California, Los Angeles, CA, USA) MACONi: a marker- and composition-aware framework for meta-Hi-C contact normalization

**P31 Jiecheng Lu** (Georgia Institute of Technology) Rethinking Sequence Models: Autoregressive Alignment and Expressive Operator Design

**P32 Muzhi Ma** (UCLA) Mechanism Design for Heterogeneous LLM Advertising

**P33 Ziling Ma** (King Abdullah University of Science and Technology (KAUST)) Adaptive Multi-Scale Forecasting and Gate-Localized Conformal Prediction for Multivariate Nonstationary Time Series

**P34 Elizabeth W. Miller** (University of Virginia) Diagnostics for Individual-Level Prediction Instability in Clinical Prediction Models

**P35 Simon Dovan Nguyen** (University of Washington, Seattle) REALITrees: Rashomon Ensemble Active Learning for Interpretable Trees

**P36 Ronan Perry** (University of Washington) Inference after selection on randomized p-values

**P37 Samya Praharaj** (Rutgers University) Bandit Simulation for Average Reward Inference

**P38 Chen Qian** (UC Davis) Structural Classification of Locally Stationary Time Series Based on Second-order Characteristics

**P39 Anik Roy** (Department of Biostatistics and Bioinformatics, Emory University) Regression Analysis of Amplitude and Phase Components in Functional Data-Based Biomarkers

**P40 Xinrui Ruan** (UC Berkeley) Learning When to Seek Human Guidance for Best LLM Judge Identification

**P41 Matias Shedden** (University of Florida) Machine-learning-based subgroup identification and adaptive enrichment in a seamless Phase II/III trial

**P42 Zhaiming Shen** (Utah State University) Understanding In-Context Learning with Kernel Methods

**P43 Zhongjie Shi** (Georgia Institute of Technology) In-Context Learning Across Tasks with Transformers: Approximation and Generalization under Low-Dimensional Structure

**P44 Shreyash Sinha** (Department of Quantitative and Computational Biology, University of Southern California) Comparative exploration of gene-based and k-mer based features for predicting antimicrobial resistance

**P45 Yang Sui** (UCLA) Statistical Inference for LLM Evaluations: From Multiple Auxiliary Predictors to Valid Uncertainty Quantification

**P46 Ao Sun** (USC Marshall) LLM-Powered Predictive Inference with Online Text Time Series

**P47 Pablo Suárez-Serrato** (UC Santa Barbara) EuLearn: a 3D database for learning Euler characteristics

**P48 Ye Tian** (Penn State University) Semi-Supervised Representation Learning with Shared Factor Structure

**P49 Yidan Tian** (Michigan State University) Minimax-Optimal Transformer Classification for Functional Data with a Dense–Sparse Phase Transition

**P50 Kaylee Vo** (Harvard University) Variational Boosting for Physics-Informed Neural Networks

**P51 Patrick Vossler** (UC San Francisco) LLMs Judging LLMs: A Simplex Perspective

**P52 Anela Leyan Wang** (UCLA, Department of Biostatistics) Nonlinear subspace factor analysis via latent bottlenecks for integrating data across studies

**P53 Kevin Wang** (Univ of North Carolina At Chapel Hill) Can Generative Artificial Intelligence Survive Data Contamination? Theoretical Guarantees under Contaminated Recursive Training

**P54 Kexin Wang** (University of Southern California) DeepLINK-T: deep learning inference for time series data using knockoffs and LSTM with applications to metagenomic data

**P55 Yizheng Wang** (Hong Kong University of Science and Technology) Unsupervised Federated Multi-task Learning for Heterogeneous Tasks

**P56 Joong-Ho Won** (Seoul National University) Pareto Variational Autoencoder

**P57 Runxiong Wu** (University of Wisconsin-Madison) VarScope-MVAE: Learning Physically Scoped Variations in Multimodal Data

**P58 Xiliu Wu** (Indiana University Bloomington) Evaluating Tabular Foundation Models for Risk Prediction: A Benchmark against Traditional Machine Learning

**P59 Qingyan Xiang** (Department of Biostatistics, Vanderbilt University Medical Center) De-biased Machine Learning for Count Data: a Partially Linear Poisson Model Based on Neural Networks

**P60 David Xie** (Department of Statistics and Applied Probability, UC Santa Barbara) Analysis of Psychological Stress Response to Physiological Data from Wearable Devices: A Study on College Student Cohort.

**P61 Tianmin Xie** (University of Southern California) Conformal Inference for Open-Set and Imbalanced Classification

**P62 Zhaoyu Xing** (University of Notre Dame) SORHA: Semi-parametric Ordered Random Hypernetwork with Attributes

**P63 Kathy Xu** (UCSB PSTAT Department) Latent Restorative Patterns Linking Physical Activity to Acute Cardiovascular Protection in Wearable Health Data

**P64 Minxing Xu** (Northwestern University) Conflict in Medical Multi-Agent Reasoning: From Disagreement to Reliable Communication

**P65 Wenlu Xu** (UCLA) Physics-Informed Kernel Learning

**P66 Haoran Yan** (Georgia Institute of Technology) Neural Scaling Laws of Transformer Operator Network

**P67 Mingsong Yan** (UC Santa Barbara) In-Context Learning for Gaussian Kernel Ridge Regression with Transformers

**P68 Tiantian Yang** (University of Idaho) Directed Weighted Graph Neural Networks with Tensor Fusion for Multi-Omics Cancer Subtype Classification

**P69 Yahong Yang** (Binghamton University) Score Estimation and Approximation of the Diffusion Model

**P70 Duoduo Danny Ying** (Department of Statistics and Data Science, UCLA) Prompt engineering using order-of-addition experiments: An application to generating two-level fractional factorial designs

**P71 Miaoqing Yu** (UC Santa Barbara) BART for multi-outcome CATE estimation with Unmeasured Confounding

**P72 Keming Zhang** (Brown University) Interpretable and Uncertainty-Calibrated Bayesian Machine Learning for Survival Outcomes with Evolving Patient Histories

**P73 Niuniu Zhang** (UCLA Anderson School of Management) Joint Calibration of LLM Agents

**P74 Sherry Zhang** (University of Texas at Austin) Code availability is not enough: schemas for documenting and extending data analysis

**P75 Yijiao Zhang** (University of Pennsylvania) Causal Network Discovery from Interventional Count Data with Latent Linear DAGs

**P76 Yuexia Zhang** (The University of Texas at San Antonio) Extracting Causal Relational Rules for Medical Question-Answering Tasks using Large Language Models

**P77 Shangcheng Zhao** (Psychology & Brain Sciences, UCSB) Moral Shadowbanning: Is Conversational AI Less Helpful, Honest, or Harmless to Users Deemed to Have Poor Moral Character?

**P78 Jialu Zhou** (University of Maryland, College Park) Source-Fair Transport Calibration for Heterogeneous AI Evaluation

**P79 Bicheng Zhu** (Department of Statistics, UC Davis) Adaptive Clustering for Heterogeneous Multiplex Networks

**P80 Jungang Zou** (Department of Biostatistics at Columbia University) AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers

Conference information: statfoundations-ai.github.io